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Beyond a Single Target: Epigenetic Intervention in Alzheimer's Disease

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RB 130

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Alzheimer's disease (AD) is a multifactorial disease in which genetic background, aging, neuroinflammation, metabolic dysfunction, synaptic failure, and decades of environmental and lifestyle exposures converge on vulnerable circuits, with pathogenic changes accumulating 15 to 20 years before clinical onset. Epigenetics is a point of convergence for these risk factors and pathogenic processes. Aging, stress, diet, sleep, and inflammation remodel the epigenome across the lifespan, and aberrant histone deacetylase (HDAC) activity silences genes required for neuroprotection, metabolic homeostasis, and repair while permitting pro-inflammatory transcription. Our laboratory targets this dysregulation with HDAC inhibitors, including M344 and the HDAC3-selective inhibitor RGFP966, and with a novel low-dose combination of RGFP966 and fractionated radiotherapy. Radiation-induced chromatin remodeling primes loci for epigenetic reprogramming, producing durable rescue of synaptic, metabolic, and neuronal gene expression in a mouse model of AD. Because this strategy acts on regulatory machinery rather than a single pathogenic protein, it is well matched to the multifactorial architecture of AD. This talk will present the supporting data and evaluate epigenetic and lifestyle interventions relative to traditional single-target strategies.